Ex. No. 2.2 – Design and Draft the Given 2D Sketches in Modelling Software

AIM

To draw the given component with the specified dimensions, create its three-dimensional model and prepare the engineering drawing using Autodesk Fusion 360.

PROCEDURE

1. Create the base plate. Start a new Fusion 360 design, set units to millimetres, and sketch an obround base on the XY plane. Set its overall length to 92 mm, width to 36 mm and thickness to 8 mm. Extrude the profile by 8 mm.

2. Add the mounting holes. On the top face, draw two circles of Ø16 mm at the rounded ends of the base. Set the distance between their centres to 56 mm and cut both holes through the plate.

3. Build the upright support and upper arm. Sketch and extrude the central rectangular support, using the 40 mm support width shown in the drawing. Add the horizontal projecting arm and position it according to the side view, which shows an overall projection of 83 mm and an overall height of 50 mm.

4. Create the inclined reinforcing web. On a side face, sketch a triangular rib joining the base plate to the underside of the upper arm. Extrude the rib across the required width so that it connects firmly to both parts.

5. Create the upper boss and bore. On the top of the projecting arm, sketch concentric circles with an outside diameter of Ø30 mm and an inside diameter of Ø20 mm. Extrude the annular profile to form the boss, then cut the Ø20 mm bore through it.

6. Finish, dimension and save. Apply the necessary coincident, concentric and symmetry constraints. Check the main dimensions: base 92 × 36 × 8 mm, mounting-hole spacing 56 mm, mounting holes Ø16 mm, support width 40 mm, and upper boss Ø30 mm with Ø20 mm bore. Compare the model with the isometric reference, then save the 3D design and create the dimensioned drawing.

SKETCHES

Reference dimensioned engineering drawing:

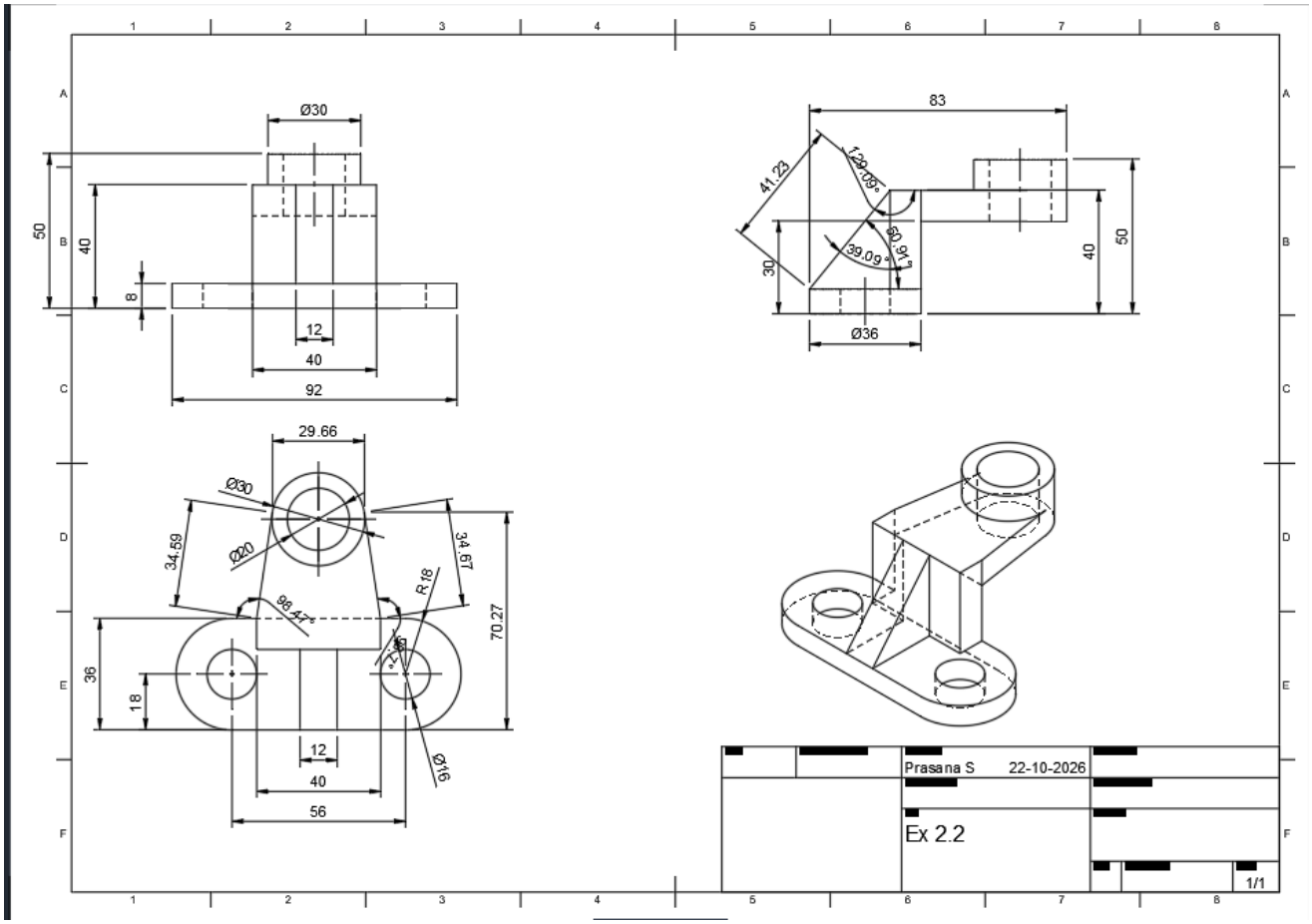


Figure 1: Dimensioned engineering drawing of the component

OUTPUT

The completed three-dimensional component modelled in Autodesk Fusion 360 is shown below.

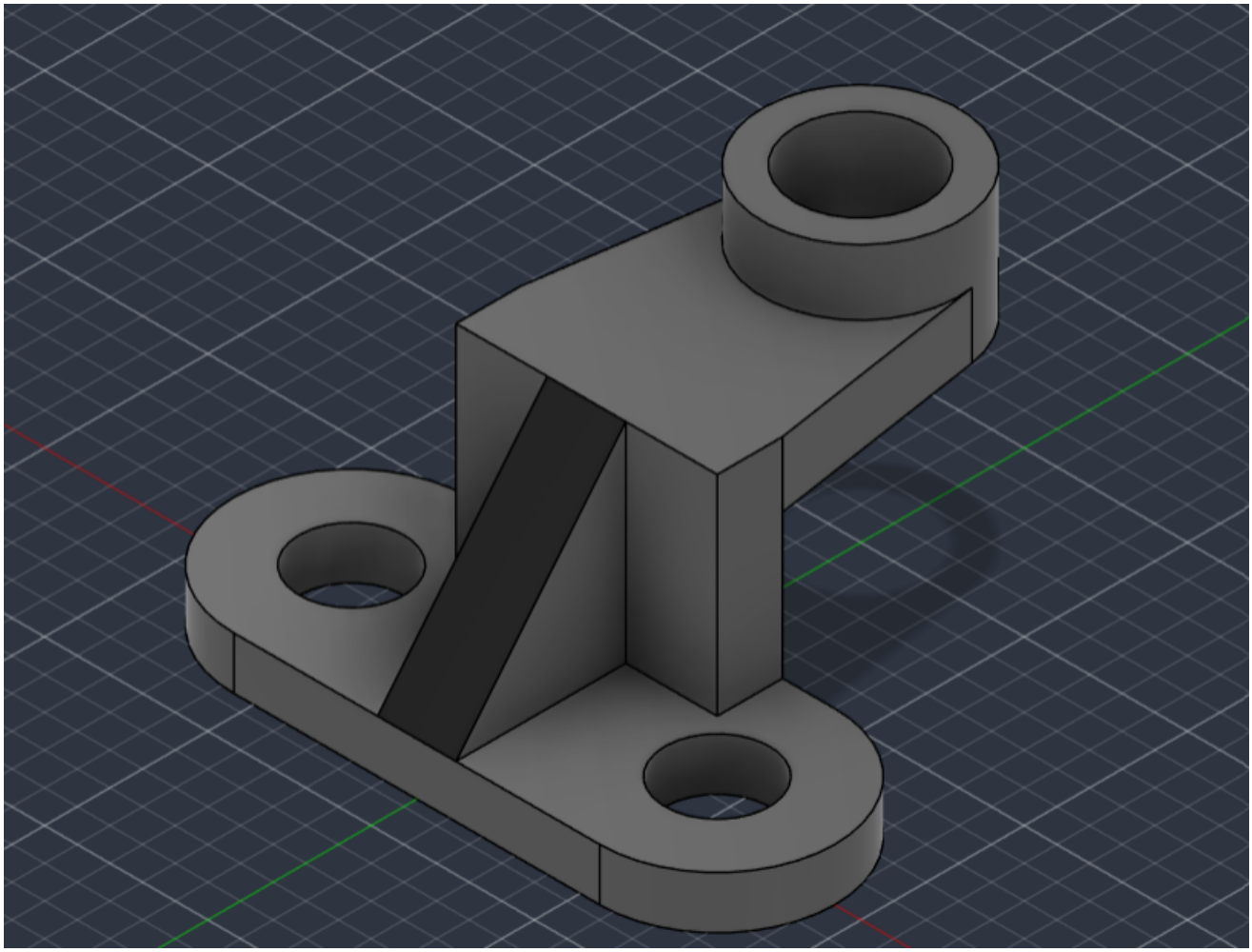


Figure 2: Completed 3D model in Fusion 360

RESULT

Thus, the given component was sketched, dimensioned, modelled and drafted using Autodesk Fusion 360.